

Sruthi Satyavarapu

M.S. Computer Science — ML Engineer | Applied Machine Learning & AI

 <mailto:sruthiraosatyavarapu@gmail.com>
 <https://linkedin.com/in/sruthi-satyavarapu>

 tel: +1 510-365-2100
 <https://github.com/sruthisDev>

 San Jose, CA
 <https://sruthisatya.wixsite.com/sruthi-portfolio>

ABOUT

Master’s Student with 4+ years of industry experience and a strong background in applied AI, conversational systems, and healthcare ML. Experienced in building end-to-end ML pipelines, RAG-based LLM applications, and scalable APIs using Python, FastAPI, and modern ML frameworks. Seeking industry ML engineering roles focused on production ML systems.

EDUCATION

M.S. Computer Science
University of the Pacific, Stockton
GPA 4.0 | Aug 2024 – Present

M.S. Technology (Medical Software)
Manipal Institute of Technology, India
GPA 3.7 | Aug 2016 – May 2018

TECHNICAL SKILLS

PythonC++Scikit-learnPyTorchLangChainHuggingFaceChromaDBTableauFastAPIPHPReactJS

MySQLMongoDBGit

RESEARCH EXPERIENCE

Research Assistant – Conversational AI (March 2025 – Present)
University of the Pacific (PI: Dr. Pramod Gupta)

- Designed modular RAG chatbot architecture with flexible integration of vector DBs, embeddings, and LLMs.
- Built ReactJS + FastAPI + ChromaDB prototype improving retrieval relevance via re-ranking and query expansion.
- Developing reusable “domain packs” for multi-domain deployment.

Research Assistant – Wearable Data Analytics (Jun 2025 – Present)
University of the Pacific (PI: Dr. Jinzhu Gao)

- Implementing ML algorithms for fall detection and sleep monitoring using accelerometer, gyroscope, and PPG data (Python, Scikit-learn, Pandas, NumPy) .
- Applying feature engineering to enhance activity classification accuracy.

PROJECTS

AI Companion for Seniors (Ongoing)
AWS Bedrock, MongoDB, FastAPI, React, Python
Working on building an AI-driven elderly-care system integrating multimodal health data, enabling real time health monitoring , anomaly detection and automated caregiver alerts.

Strawberry Ripeness Detection (2024)
Python, Scikit-learn, Pandas
Built an ensemble classifier using a bagging-style combination of KNN and SVM, achieving 92% accuracy for strawberry ripeness prediction

Weather–Music Trends Analysis (2024)
Python, Seaborn, Tableau
Performed exploratory data analysis across 100+ cities to identify correlations between weather conditions and music listening patterns, uncovering genre–climate trends using top charts

Adidas Sales Dashboard (2024)
Python, Tableau
Built an interactive dashboard to visualize regional sales performance, product trends, and key business KPIs for decision support.

INDUSTRY EXPERIENCE

Developer – Tata Consultancy Services, Hyderabad (Jul 2021 – Jul 2023)
Contractor (Jul 2021 – Oct 2022), Full-time (Oct 2022 – Mar 2023)

- Developed and managed critical B2B applications for MARS PetCare, improving backend performance by 20
- Mentored new team members and led onboarding, providing hands-on guidance and technical training.

Technologies: Ez, MySQL, PHP, JavaScript

Web Developer / Software Engineer – Hyper Interact, Foray, PurpleTalk, Excellera (2018 – 2021, India)

- Enhanced web applications and internal tools across multiple projects, improving efficiency and user experience (e.g., 30
- Developed mobile and web solutions in bioinformatics, e-commerce, and digital services; received multiple performance awards.
- Streamlined workflows via automation and feature optimization, increasing productivity by up to 25%.

Technologies: PHP, Drupal, Yii, MySQL, MongoDB, JavaScript, JQuery, HTML/CSS